Prepared designs / what fits now

F005 / evidence R004 / DRAFT - NOT FOR FABRICATION

Two coordinated pairs are prepared for prototype and quote review. Both preserve the primary scan and reuse the original Kohler trim. The current hardware cannot be claimed to fit unchanged.

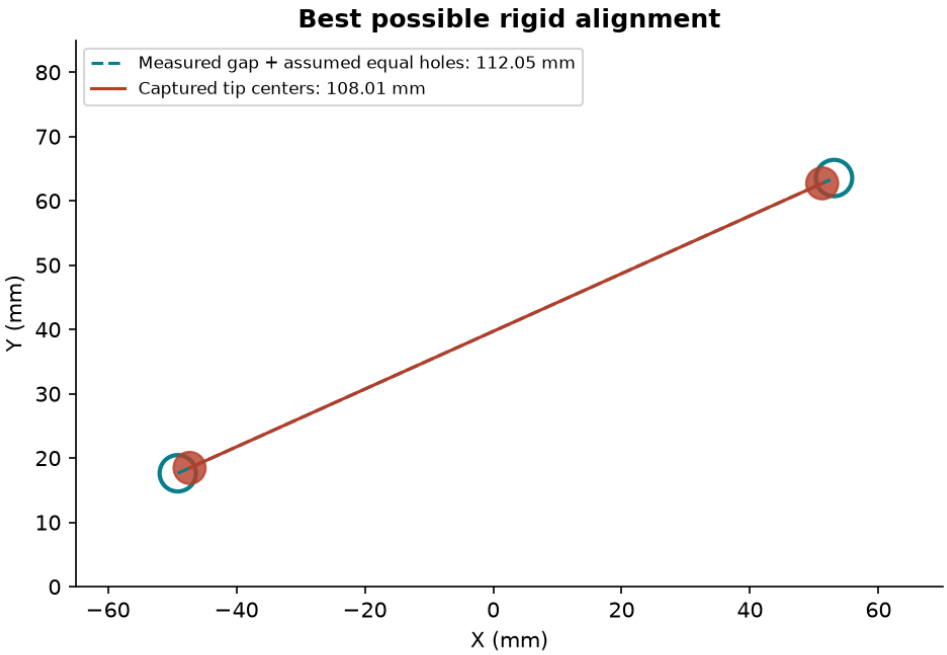


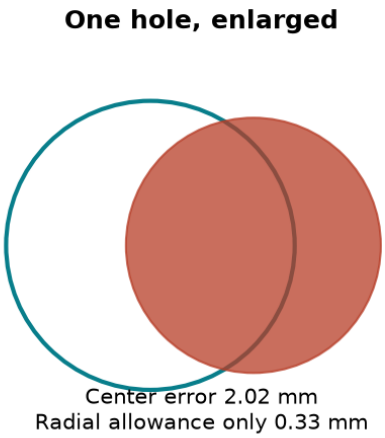


Design	Main / diverter	Projected gap
A - round	230 mm OD / 120 mm OD	6.24 mm
B - rounded rectangle	220 x 218 R62 / 120 x 120 R35	11.24 mm

Appearance views use the actual travertine and official Kohler exterior meshes. Captured hardware is hidden only in these cosmetic views so the intended appearance can be assessed; the unchanged observed state and conflicts remain in Blender. Neither image is evidence that the cover currently seats.

All profiles use a new shaped main aperture, not the rejected 118 x 136 mm rectangle. The separate diverter opening is an 80 mm trial and requires its factory rear-plate/foam check.





Conditional conflict: actual stud axes/diameters, NE hole diameter and scan scale remain unresolved.

1. Diagonal mounting pattern

106.39 mm nearest-edge gap + 5.66 mm, assuming equal small holes, gives 112.05 mm centers. The captured tip centers are 108.01 mm apart. Even the best rigid alignment leaves 2.02 mm center error per hole versus only 0.33 mm radial space around an assumed 5 mm stud. This is a model conflict; reflective tip picking, stud tilt, NE hole size and scan scale can change the real result.

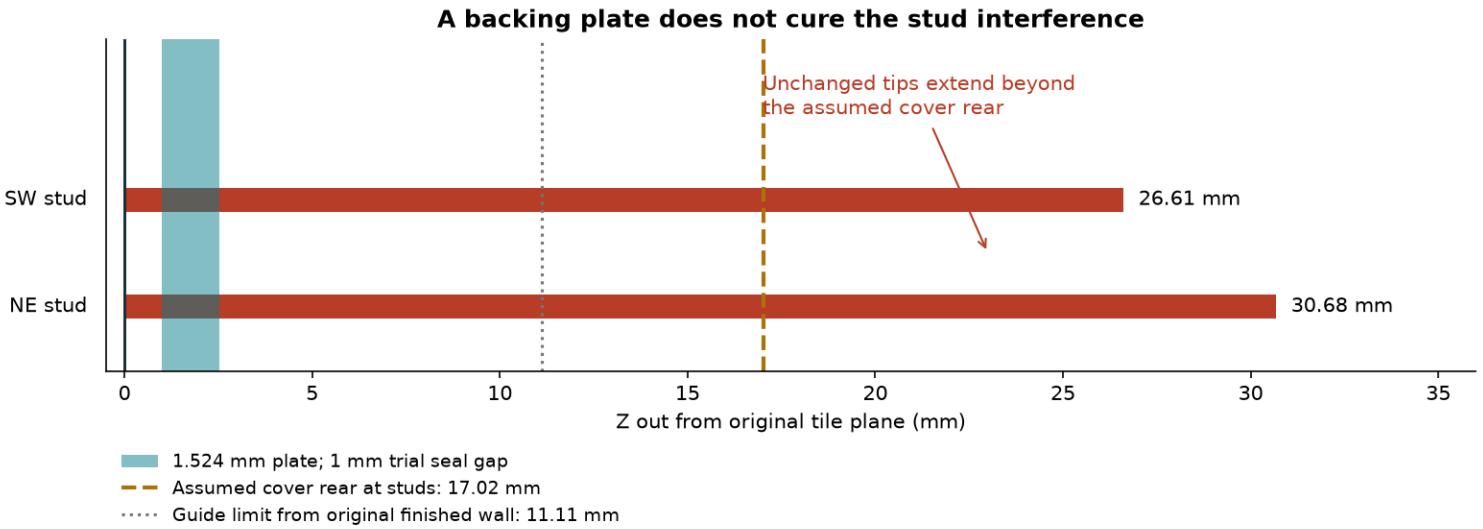
2. Outer-cover controls

The inside-edge interpretation of 46.25 mm with two 33.79 mm bores gives 80.04 mm spacing, 0.665 mm above the 79.375 mm nominal anchor. The lower bore is unmeasured. The reported 35.23 mm lower-rim gap then puts the lower center 52.125 mm above the cover bottom. This is a plausible fit scenario, not an accepted registration.

3. Actual control surfaces versus old cylinders

At Z about 14 mm the captured lower control surface within a 33 mm radius reaches roughly 17.1 mm radius; at Z about 18 mm it reaches about 14.2 mm. The old 30 mm-radius straight cylinder overstated that envelope. The lower bore, local cover profile and adapter/bonnet engagement still need dry-fit confirmation.

A separate exact inside-solid test also finds captured points inside the provisional inner plate (149-164 points over four trial foam depths) and cover (202 points). All detected points are in stud neighborhoods, with none in control neighborhoods at these trial poses. These are reference-shape conflicts, not proof of defective real parts: the stamped seats, rear cavity, foam compression and registration are unmeasured.



The studs still block the cover in the depth model

With the thin metal and a 1 mm illustrative gap, the assumed cover rear is at Z=17.02 mm. Stud tips at 26.61 and 30.68 mm exceed it by about 9.59 and 13.65 mm. Even the 3.175 mm metal option does not eliminate that conflict. Do not convert these numbers into cutting instructions: actual local cavities, axes, adjustment and thread engagement are unknown.

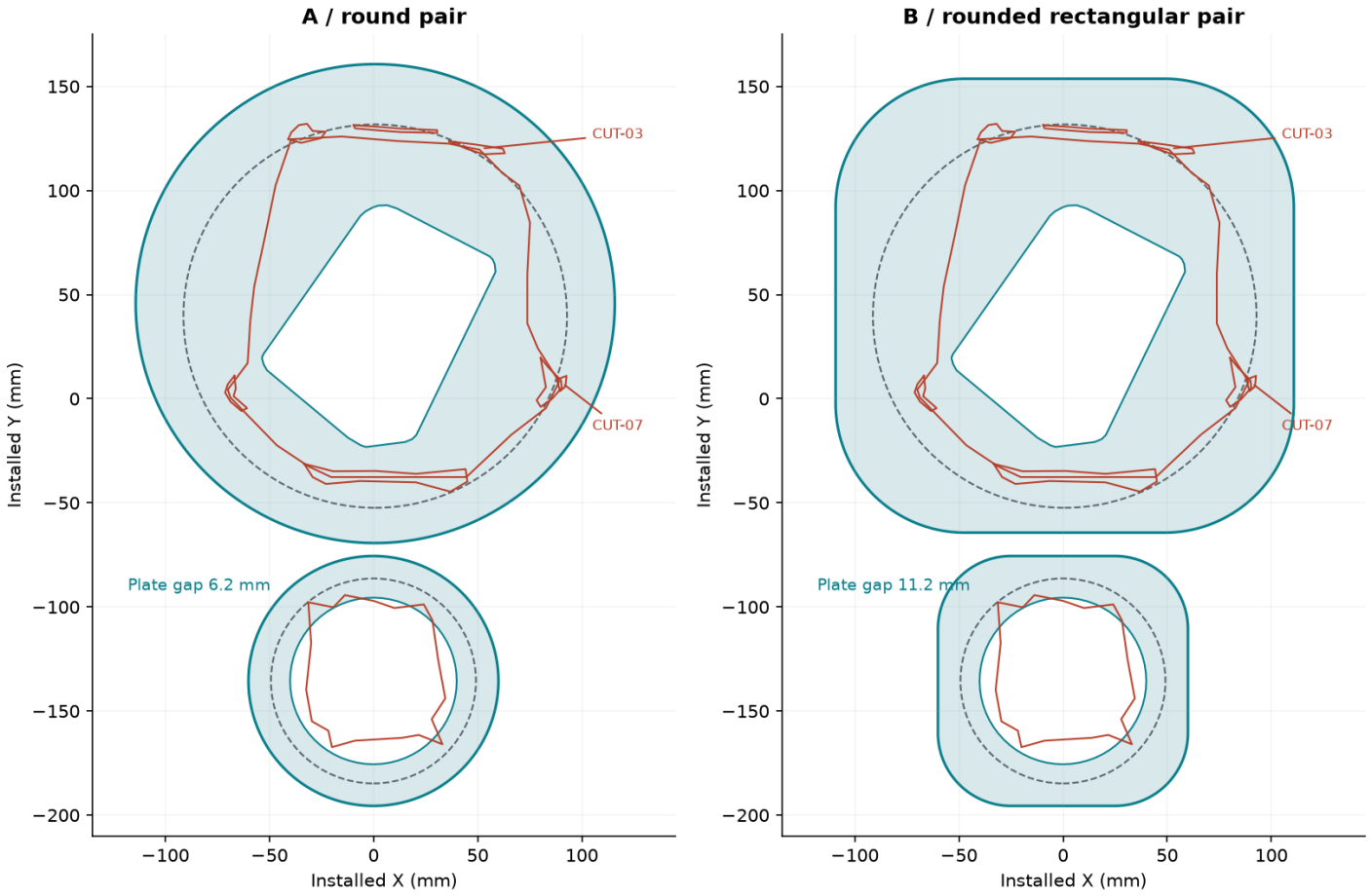
The new aperture needs the actual rear foam trace

If the 6 mm foam band follows the outer metal outline, the trial leaves about 1.45 mm from aperture to the foam inner edge. Moving that assumed path inward by 1.5 mm creates a small intersection. The former 116 mm-wide capsule placeholder is not the actual measured gasket. A back-face trace against the supplied template is more useful than guessing another diameter.

Handle/stem projection alone does not diagnose shower-water entry. Cover seating, the intended factory gasket path, supported contact and unobstructed drainage must work together. The free plate depth does not establish an allowed installed compression. No general 5 mm waterproofing allowance is justified.

Coverage / insertion / support

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Red: traced damage. Dashed: factory trim. Blue: custom material. All scan distances remain conditionally scaled.

Profile	Limiting trace	Margin after 1.5 mm trace allowance
MAIN-A	CUT-07	13.87 mm
MAIN-B	CUT-07	16.64 mm

The forward-insertion check considers 51,929 full-resolution clipped triangles inside the eroded wall opening: none projects beyond the new aperture. Minimum sampled clearance is 1.55 mm. The boundary mask, hidden surfaces and unbounded scale prevent a physical clearance pass.

No structural load or clamp force is known. Relative stiffness of 3.175 versus 1.524 mm stainless is about 9.04x. That comparison does not prove either can bridge the actual unsupported span. Check bearing, rocking, local flex and the grout crossing before choosing thickness.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

USD planning allowances for one main plate plus one matching diverter plate, quantity one each. Ranges are assistant estimates, not published part prices or vendor quotes. No tax, shipping upgrade, installation, primer/sealant or separate cosmetic finishing included unless noted. Exact vendor totals require a later authorized quote using the chosen geometry and finish.

SendCutSend / Clear acrylic laser prototype, 3.0 mm

Planning pair allowance: $40-100. No price for this pair. Pricing depends on material, geometry, quantity and processing; published free US shipping threshold is $39.
Standard sheet cutting ships in 2-4 business days. 0.118 inch acrylic appears on rush eligibility list; 24-hour production is conditional on checkout eligibility and noon PT cutoff.
Keep protective film; verify actual thickness. Prototype only; add a measured 0.175 mm spacer to imitate the 3.175 mm metal stack, if needed.

Source: https://sendcutsend.com/materials/

Ponoko / Clear acrylic laser prototype, stock near 3 mm

Planning pair allowance: $50-130. Advertises acrylic parts from $50 for one; that is a starting offer, not a price for these plates.
Advertises same-day production/delivery options. Rush order cutoff 11 AM; same-day local Bay Area or next-day mainland-US options are conditional. Confirm material, destination and current cart date.
Clear acrylic exposes the foam landing and valve interference during dry fit. Custom non-stock materials add fees/delay, so use standard stock.

Source: https://www.ponoko.com/laser-cutting/acrylic

3D Printer on Demand / FDM plastic dry-fit pair

Planning pair allowance: $40-120. No fixed design price. Vendor states no tooling, setup fee or minimum order on its rapid-prototyping page.
Published typical production/ship lead time 3-5 business days; transit additional.
PLA/PETG published instant-pricing size limit 250 mm; 230 mm plate fits that stated envelope. Ask for a single flat print, no scaling, and thickness/flatness inspection. Thin wide prints can warp. Published domestic shipping $7.95 per order, additional to the planning range.

Source: https://www.3dprinterondemand.com/use-cases/rapid-prototyping

Xometry / Industrial FDM plastic prototype pair

Planning pair allowance: $100-300. Design-specific quote required; reserve range is an assistant budget, not a published vendor rate.
US 3D-printing page advertises 1-day FDM expedites. Standard and actual delivery date depend on geometry/material/capacity; eligible domestic 3D printing has free 2-day or ground shipping.
Use only if a confirmed expedited date justifies its quote. Check flatness, thickness and print orientation; a smooth appearance alone does not establish fit.

Source: https://www.xometry.com/capabilities/3d-printing-service/

Metal cost / lead-time research

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Prices below are unquoted planning reserves for one pair. Cosmetic finishing, taxes and shipping upgrades may add cost. Published production time is not a guaranteed delivered date.

SendCutSend / 316 stainless laser pair, 0.060 or 0.125 inch

Planning pair allowance: $65-180. No part-specific price. Published material options include 0.060 and 0.125 inch; laser tolerance +/-0.005 inch (0.127 mm).
Standard ships in 2-4 business days; deburring adds 0 days, detabbing +1 day. The current 316 rush list contains 0.187 and 0.250 inch ONLY, so do not assume 24-hour service for these thin plates.
Request deburring and tab removal. Raw/deburred stainless is not Kohler BN. Reserve an additional $50-150 and 2-5 business days for separate cosmetic brushing only as an unquoted planning contingency.

Source: https://sendcutsend.com/materials/stainless-steel-316/

OSH Cut / 304 #4 brushed stainless 0.060 inch, or 316 #2B 0.060 inch

Planning pair allowance: $75-220. No geometry-specific price. Official materials list includes 304 #4 brushed with protective film and 316 #2B in 0.060 inch. No minimum order amount advertised.
Standard laser-cut prototype/short-run lead time 2 business days; same-day options depend on job size, stock and capacity. Confirm extra time for selected finishing and transit.
304 #4 stock offers the simpler cosmetic route. 316 has greater chloride resistance but its listed #2B surface is not a matching brush. Deburring/graining offered; request finish protection, grain direction and edge treatment.

Source: https://www.oshcut.com/

Ponoko / 304 stainless laser-cut pair

Planning pair allowance: $90-250. No part-specific price obtained; material page confirms 304 options.
Current service page advertises same-day cutting; verify stock/finish eligibility. Do not use the older 2019 blog's 15-day lead time as today's quote.
Ask for stock thickness matching the chosen STEP and a protected cosmetic surface. Quoted alloy, finish, tolerance and delivered date must be reconciled before release.

Source: https://www.ponoko.com/laser-cutting/stainless-steel

Best next purchase path

First dry-fit the clear 3 mm acrylic pair using the same DXF. This exposes the critical narrow foam-to-aperture margin and costs less than remaking a finished metal pair. If direct-to-metal is chosen after fit reconciliation, compare 316 0.060 inch for low depth and 0.125 inch for stiffness; compare OSH Cut 304 #4 stock when a ready brushed cosmetic surface is more important. Exact Kohler BN color is not promised.

Quote / prototype review only. Physical fit, seal support and manufacturing release unresolved.

Selected-system work still required

316 stainless preferred corrosion-resistance candidate; 304 brushed is a viable comparison, subject to environment/cleaner and finish acceptance. No coating or plating is assumed to match BN.

3 mm STL supplied. Dry fit only: no water-sealing, structural or long-term wet-service approval of FDM/PMMA. A 230 mm one-piece main plate needs a print bed with usable width at least 230 mm plus brim allowance; a nominal 220 mm bed is too small.

LATASIL neutral-cure silicone is a research candidate for stone/metal interfaces. Its data sheet specifies LATASIL 9118 Primer for stainless steel and porous stone, subject to testing/approval. Verify adhesion and nonstaining on this travertine. No generic 1 mm full-face adhesive layer or minimum foam compression is approved here.

LATASIL data sheet states protect from water until fully cured, generally 24 hours at 70 F / 21 C. Actual joint section, temperature, humidity, substrate and other repair materials can change the wait. This is not an installation release or a cure schedule for an unselected patch system.

LATASIL DS-6200.1-0926 also requires a test area on stone; it describes joints at least 3 mm wide and approximately 2:1 width/depth for perimeter joints. It does not validate our illustrated 1 mm lap layer. A stone/metal seal detail must be reviewed before permanent installation. Factory foam and drainage must remain functional.

Four focused physical checks to release a design

[ ] Rear foam: trace both edges on the full-size main template; inspect the diverter factory rear assembly too.

[ ] Hardware: remeasure the two diagonal studs and identify the rear contact; dry-fit the real inner plate, cover and handles without forcing.

[ ] On-wall prototype: confirm coverage, exact trim registration, gap between plates, flat support/rocking, full handle travel and removal access.

[ ] Final choice: select A or B, thickness, alloy/finish, validated support/seal detail and vendor quote; record explicit manufacturing release.

These checks replace repeated broad measurement requests. The accepted caliper readings are preserved; only the missing fit-critical information is needed next.

Source register / scope of verification

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Public primary sources reviewed 2026-10-14. Original photos, raw measurements, scan registration and the complete original goal remain in the private repository. No vendor contact, CAD upload, order, physical dry fit or water test was performed.

https://sendcutsend.com/materials/

https://sendcutsend.com/pricing/

https://sendcutsend.com/sendcutsend-processing-times/

https://www.ponoko.com/laser-cutting/acrylic

https://www.ponoko.com/laser-cutting

https://help.ponoko.com/en/articles/4468203-custom-material-orders

https://www.3dprinterondemand.com/use-cases/rapid-prototyping

https://3dprinterondemand.com/

https://3dprinterondemand.com/pricing

https://www.xometry.com/capabilities/3d-printing-service/

https://community.xometry.com/kb/articles/651-how-much-is-shipping

https://community.xometry.com/kb/articles/642-what-is-your-minimum-order-quantity

https://sendcutsend.com/materials/stainless-steel-316/

https://www.oshcut.com/

https://www.oshcut.com/services/deburring

https://www.ponoko.com/laser-cutting/stainless-steel

https://www.ponoko.com/sheet-metal-fabrication

https://techcomm.kohler.com/techcomm/cad/T14489-4.obj

https://techcomm.kohler.com/techcomm/cad/T14491-4.obj

https://techcomm.kohler.com/techcomm/pdf/1216755-2.pdf

https://resources.kohler.com/onlinecatalog/pdf/1041504_2.pdf

https://techcomm.kohler.com/techcomm/pdf/1190382-2.pdf

https://cdnmdm.laticrete.com/Datasheet/NA/101/Datasheet_NA_en_101.pdf

Software verification includes valid single solids, STEP reimport dimensions/volume, STL watertightness, DXF reimport and two-contour checks, non-fused assemblies, captured triangle clearance, and saved Blender reopen/dimensions. These checks do not validate concealed waterproofing or installed sealing.